

Predicting Super Bowl 2026 Attendance

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Data Background

- **Objective**: The goal of this project is to build a regression model that predicts Super Bowl attendance on a combination of team performance, ticket price, entertainment factors, venue characteristics, attendance, and host-city and state.
 - The aim is to identify which factors most strongly influence attendance levels across different Super Bowls and attempt to predict attendance for the 2026 Super Bowl

Data Retrieval

Variables:

- **Year** – Web scraped [NFL Playoff Pass](#)
- **Teams (Winner and Loser)** – Web scraped - NFL Playoff Pass
- **Stadium** TopEnd Sports
- **Indoor/Outdoor** Wikipedia
- **Stadium Capacity** Wikipedia
- **Attendance** Statista
- **Average Ticket Price**- Yahoo Sports
- **City, State** TopEnd Sports
- **Type of Halftime Performance** - ESPN
- **Regular Season Win Percentage (Winner and Loser)** – Pro Football Reference

Data Characteristics

Field types

- **Numeric (continuous):** Attendance, Ticket Price, Winner RS Win %, Loser RS Win%, Year
- **Categorical (strings):** Team 1, Team 2, Stadium, City, State, Super_Bowl (Roman).
- **Booleans / binaries:** Indoor (0 = Outdoor, 1 = Indoor).
- **Ordinal category:** Type of Halftime Performance *(1/2/3; see mapping below)

* Halftime Performance (1/2/3):

1 = marching/themed or non-marquee act

2 = mid-level/mixed artists

3 = superstar headliner

Key Assumptions, Risks, and Current Decisions

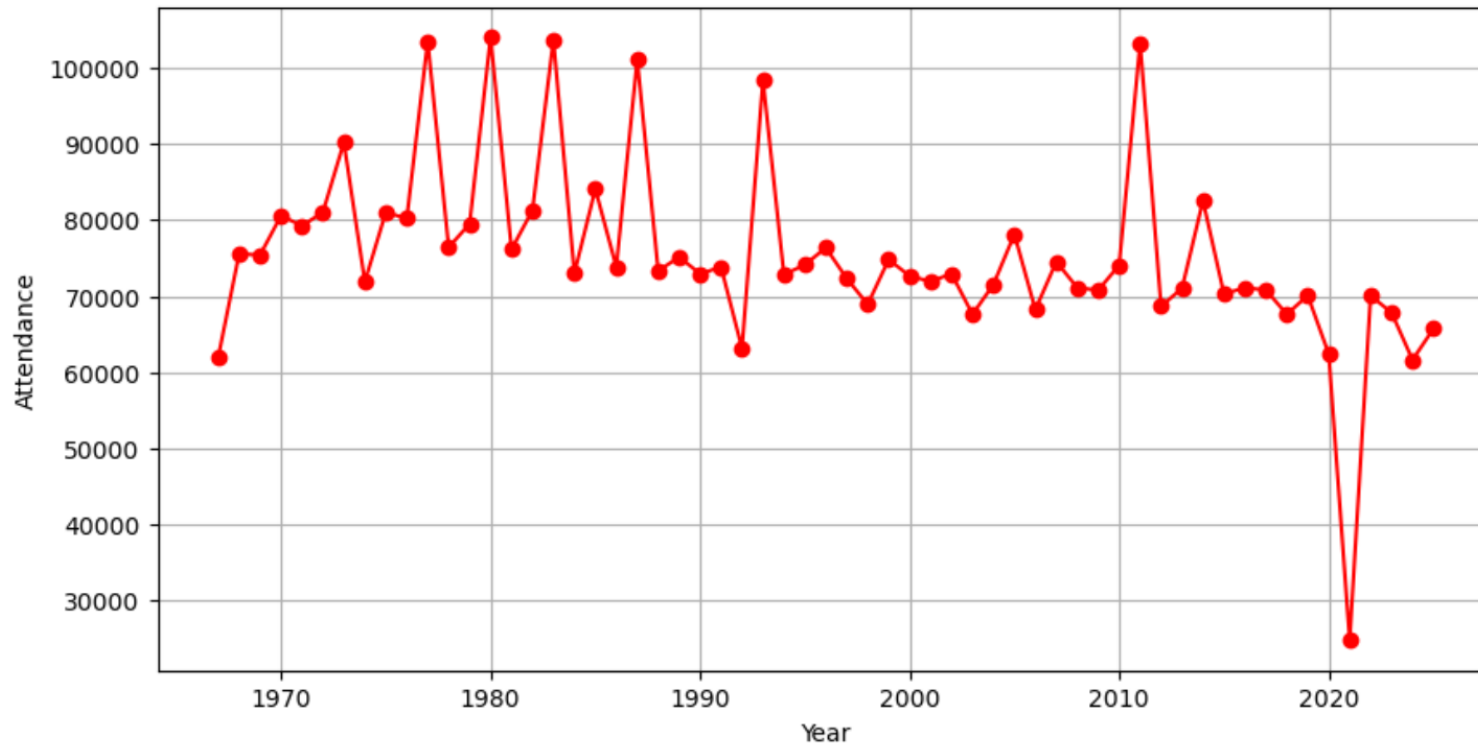
- COVID year (2020/2021) flagged for sensitivity analysis to handle extreme outlier effects
- Binary classification of indoor/outdoor
- 1,2,3 performance classification
- Regular season notes (players strikes, changes, etc.)
- Capacity
- Stadium/team/franchise evolution (stadiums being rebuilt/renamed, teams being renamed)

	Year	Team 1	Team 2	Ticket Price	Super_Bowl	Stadium_x	Indoor	City	State	Type of Halftime Performance	Attendance	Winner RS Win %	Loser RS Win %	Stadium_y	Capacity
0	2025	Philadelphia Eagles	Kansas City Chiefs	6304	LIX	Caesars Superdome	1	New Orleans	Louisiana	3	65719	0.824	0.882	Caesars Superdome	73208
1	2024	Kansas City Chiefs	San Francisco 49ers	9400	LVIII	Allegiant Stadium	1	Paradise	Nevada	3	61629	0.647	0.706	Allegiant Stadium	65000
2	2023	Kansas City Chiefs	Philadelphia Eagles	9915	LVII	State Farm Stadium	1	Glendale	Arizona	3	67827	0.824	0.824	State Farm Stadium	63400
3	2022	Los Angeles Rams	Cincinnati Bengals	7654	LVI	SoFi Stadium	1	Inglewood	California	2	70048	0.706	0.588	SoFi Stadium	70240
4	2021	Tampa Bay Buccaneers	Kansas City Chiefs	6173	LV	Raymond James Stadium	0	Tampa	Florida	3	24835	0.688	0.875	Raymond James Stadium	69218

Note: Some stadiums have the capacity to expand their seating for special occasions. Thus, some Super Bowls' attendance will be larger than the Stadium capacity

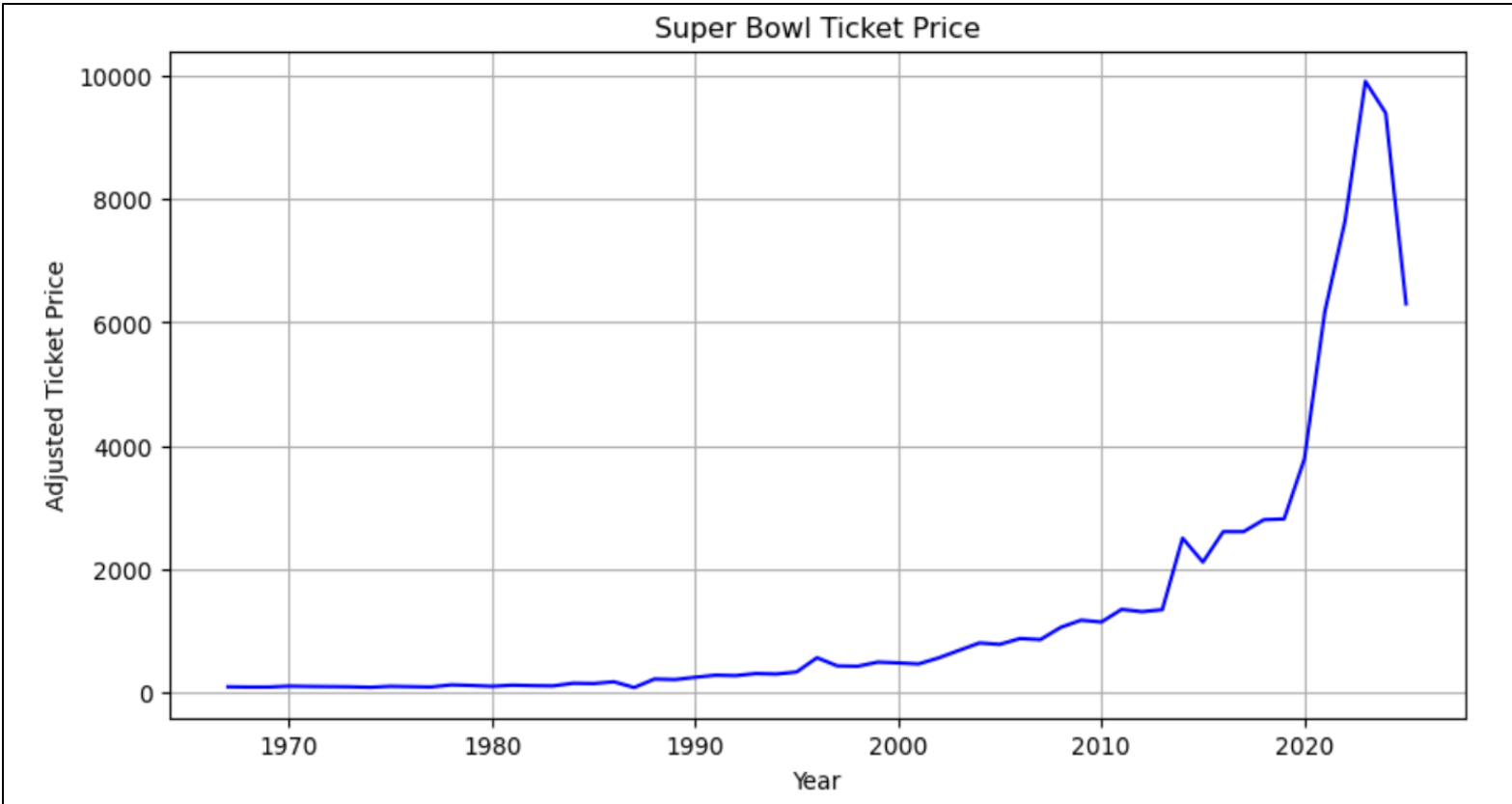
EDA (Attendance)

Super Bowl Attendance Through the Years



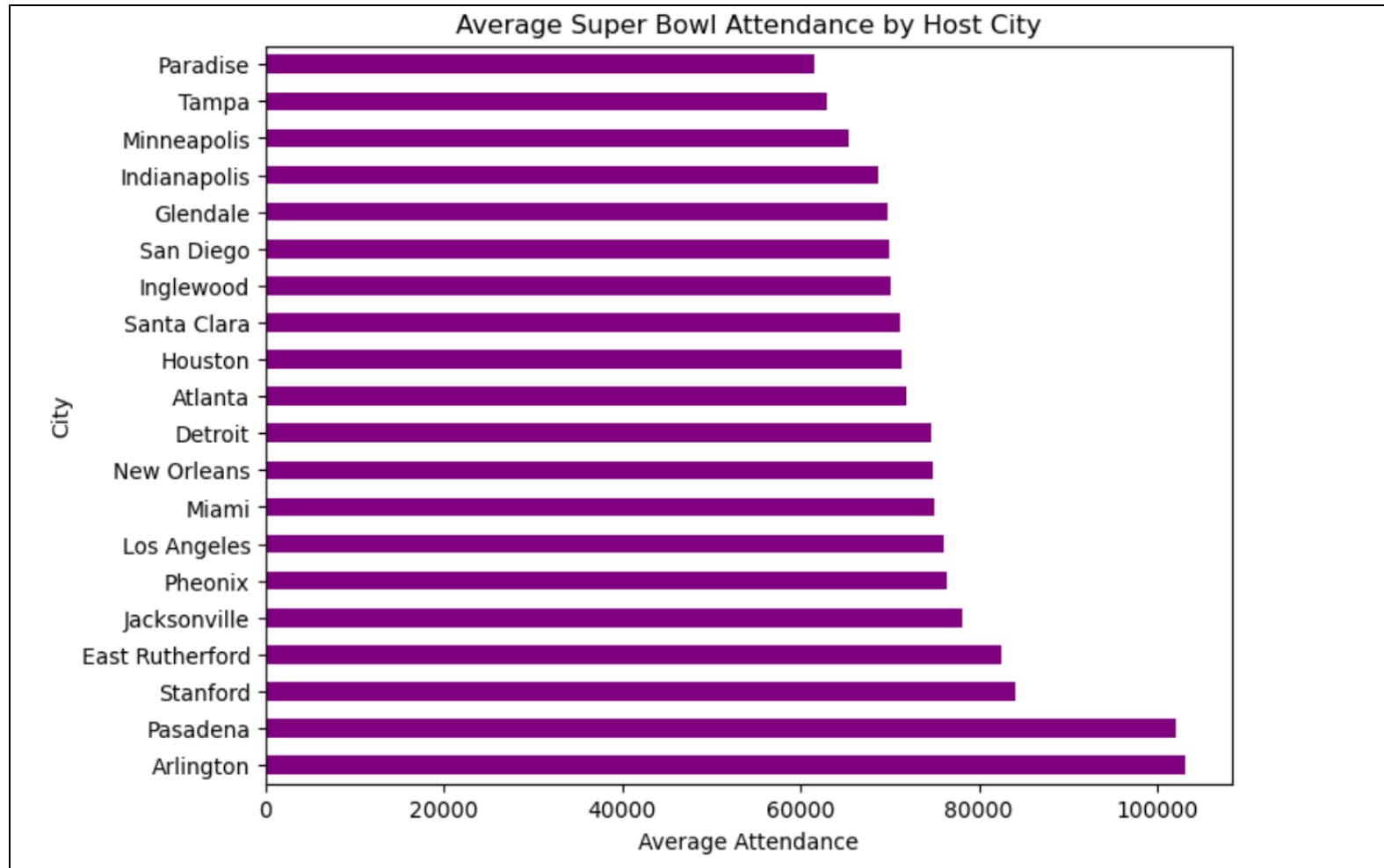
- This graph depicts the attendance to all the Super Bowls through the years
- The graph mainly fluctuates between 60,000 and around 100,000.
- See major fluctuations in the early years of the Super Bowl, but steadies starting around the late 1990s (see some outliers post-2010).

EDA (Ticket Price)



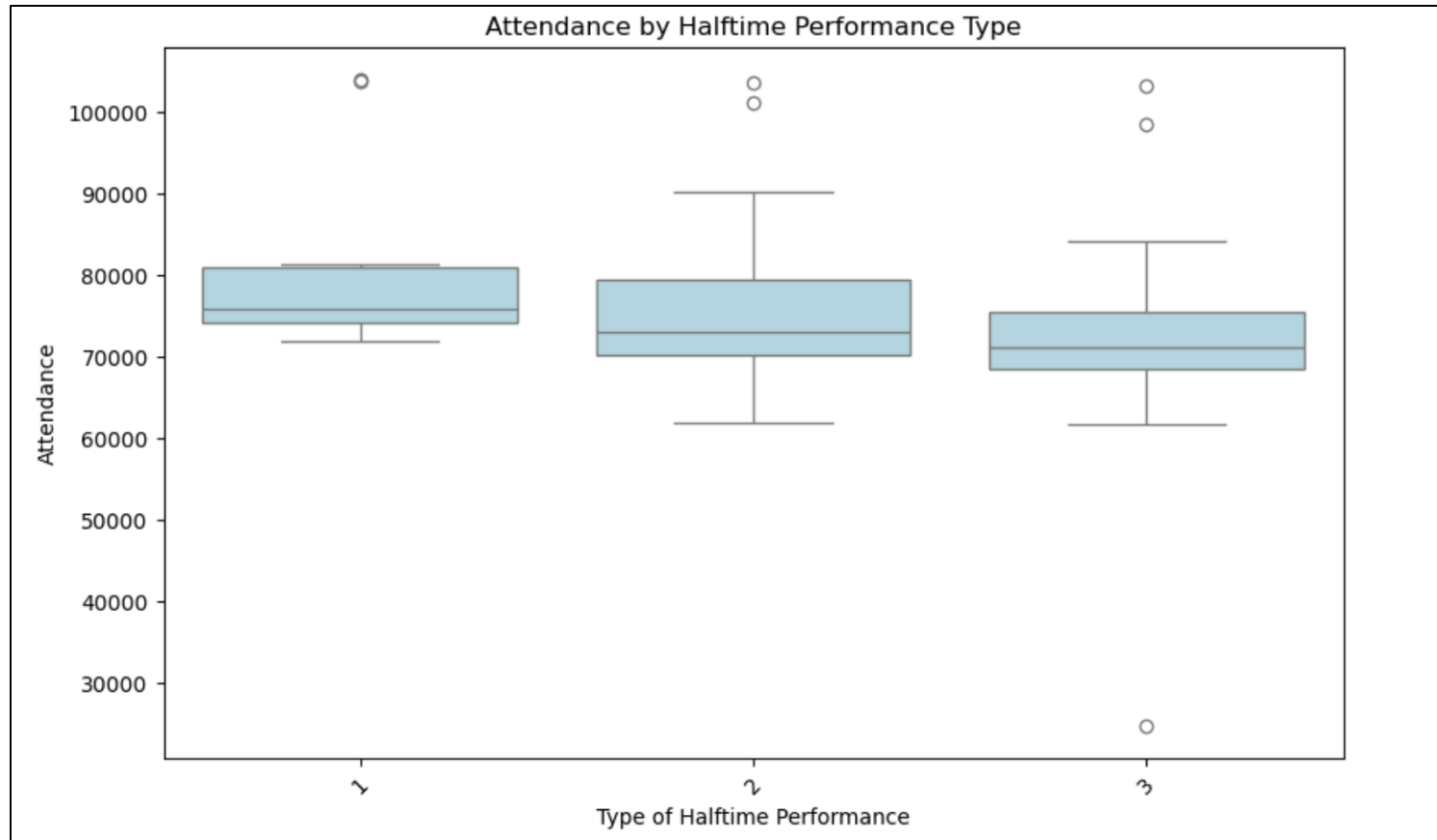
- The graph to the left depicts the ticket price for the Super Bowl through the years.
- Until around 2010, the ticket price remained steady.
- We see a huge spike around 2021 and 2022 due to COVID-19 and the regulations surrounding the pandemic.
- Since COVID, the ticket prices have dropped slightly.

EDA (Attendance by Host City)



- The chart shows the average attendance for the Super Bowls by host city.
- The attendance tends to be larger when the Super Bowl is hosted in Arlington, Pasadena and Stanford.

EDA (Attendance and Halftime)



- The box-and-whisker plot shows that the median attendance is around the same for the 3 types of halftime performances.
- The box and whisker plot is the tightest for Type 1 indicating the attendance to be consistent.
- The plots are longer for types 2 and 3 showing more fluctuation.

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